MATH 323: Probability

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Abstract

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1 Introduction

These notes were written for MATH 323 (Probability) at McGill University, Summer 2026. The course is taught by Magid Sabbagh and meets in Steward Biology Building S1/3 (S1/3) every Monday, Tuesday, Wednesday, and Thursday from 8:35 am to 10:55 am.

Grading scheme:

$$20\% \text{ assignments} + 5\%\text{Multiple Choice Quiz} + 25\% \text{ midterm} + 50\% \text{ final} \quad (1)$$
$$\text{or} \quad (2)$$
$$20\% \text{ assignments} + 5\%\text{Multiple Choice Quiz} + 75\% \text{ final} \quad (3)$$

whichever is higher. The book used for this class is: "Mathematical statistics with application, 7th edition." Midterm will be an 1h30 long and will have all 2h30 hours to complete it in class.

2 Prerequisite Knowledge

May 4, 2026

Basic Set Theory:

- A set is a well defined collection of objects, defined in a way such that we can determine wether or not an element x belongs to the set.
- If A is a set, we say that $a \in A$, if a is an element of A.
- If A is a set, we say that $a \notin A$ if a is not an element of A.
- We specify a set either by listing its elements, if possible on by a property that defines the set.
- In general, we will be working in a universal set Ω.
- Universal set is a set in which all elements belong, like the complex set.
- Operations on sets, let Ω be a universal set, $A, B \subseteq \Omega$. Then we can define the following operations on sets:
  - Subset: $A \subseteq B$ if $x \in A$ implies $x \in B$.
  - Equality: $A = B$ if $A \subseteq B$ and $B \subseteq A$.
  - Proper subset: $A \subsetneq B$ if $A \subseteq B$ and $A \neq B$.
  - Union: $A \cup B = \{x : x \in A \text{ or } x \in B\}$
  - Intersection: $A \cap B = \{x : x \in A \text{ and } x \in B\}$
  - Difference: $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$
  - Complement: $A^c = \{x : x \in \Omega \text{ and } x \notin A\}$

- Singleton: $\{x\} = \{y : y = x\}$, the set that contains only the element x.

The complement of $A \subseteq \Omega$ is denoted by A^c or $\overline{A}$ and is defined to be $A^c = \{x \in \Omega : x \notin A\}$. The empty set is denoted by $\emptyset$.

Proposition 1. *$A \cup A^c = \Omega$, $A \cap A^c = \emptyset$, $A \cup \emptyset = A$, $A \cap \emptyset =$.*

Proposition 2. *Let A,B,C be sets:*

1. *Commutativity: $A \cup B = B \cup A$.*
2. *Associativity: $A \cup (B \cup C) = (A \cup B) \cup C$, $A \cap (B \cap C) = (A \cap B) \cap C$.*
3. *Distributivity: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.*

Proof. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, $A \cap (B \cup C) \subset (A \cap B) \cup (A \cap C)$. □

Let $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$. Then $x \in A$ and x belongs to atleast B and C. If $x \in A$ and $x \in B$ then $x \in A \cap (B \cup C)$? If $x \in A$ and $x \in C$, then $A \cap C \subset (A \cap B) \cup (A \cap C)$. Hence $A \cap (B \cup C) \subset (A \cap C)$. We now show that $A \cap B) \cup (A \cup C) \subset A \cap (B \cup C)$. Let $x \in (A \cap B) \cup (A \cap C)$. If $x \in A \cap B$ then $x \in A$ and $x \in B \subseteq B \cup C$. If $x \in A \cap C$, then $x \in A$ and $x \in C \subseteq B \cup C$. So $x \in A \cap (B \cup C)$. Hence $(A \cap B) \cup (A \cap C) \subset A \cap (B \cup C)$. De morgans law:

1. $(A \cup B)^c = A^c \cap B^c$.
2. $(A \cap B)^c = A^c \cup B^c$.

Let $A_1, A_2, A_3, \ldots$ be subsets of Ω. Let $n \in \mathbb{N} : \cap_{i=1}^n A_i = A_1 \cap A_2 \cap \ldots \cap A_n = \{x \in \Omega : x \in A_1, \ldots, x \in A_n\}$. $\cap_{i=1}^\infty A_i = \{x \in \Omega : x \in A_i \forall i \in \mathbb{N}\}$. $\cup_{i=1}^\infty A_i = \{x \in \Omega : x \in A_{i_0}$ for some $i_0 \in \{1, \ldots, n\}\} = \{x \in \Omega : x \in$ at least one of $A_1, \ldots, A_n\}$.

Proposition 3. *$(\cup_{i=1}^\infty A_i)^c = \cap_{i=1}^\infty A_i^c$ and $(\cap_{i=1}^\infty A_i)^c = \cup_{i=1}^\infty A_i^c$.*

Problem 1. *In a class of 101 students. 20 students play hockey. 30 students like math. 5 students like math and play hockey. How many students, (1) do not play hockey, (2) do not like math, (3) do not like math and do not play hockey, (4) play hockey but do not like math. Answer (1): $101 - 20 = 81$ students do not play hockey. (2) $101 - 30 = 71$ students do not like math. (3) $101 - (20 + 30 - 5) = 56$ students do not like math and do not play hockey. (4) $20 - 5 = 15$ students play hockey but do not like math.*

We define $A \setminus B = \{x \in \Omega : x \in A and x \notin B\} = A \cap B^c = A \setminus (A \cap B)$.

Remark 1. *Write $|A|$ (cardinal of A) for the number of elements in A. Then $|A \cup B| = |A| + |B| - |A \cap B|$.*

Proof. $|A \cup B| = x + y + z, |A| = x + y, |B| = y + z, |A \cap B| = y$. Then $|A \cup B| = |A| + |B| - |A \cap B|$. $\square$

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Example. *1) Let Ω be a set. $\mathbb{F} = \{\phi, \Omega\}$ is a σ-algebra on Ω.*
2) Let Ω be a set. Then the set of all subset of Ω is a σ-algebra on Ω. (The power set of Ω denoted by Z^{Ω}).
3) Let ω be a set. Define $\mathbb{F} = \{A \subseteq \Omega. A$ is countable on $A^c$is countable$\}$. Then $\mathbb{F}$ is a σ-algebra on Ω. 4) Suppose $A \subseteq \Omega$. $\mathbb{F} = \{\phi, A, A^c, \Omega\}$ is a σ-algebra.

Example. *Flip a coin. $\Omega = \{H, T\}$. $2^n = \{\phi, \{H\}, \{T\}, \{H, T\}\}$.*

Definition 1. *Let Ω be the sample space (...) to an experiment and $\mathbb{F}$ a σ-algebra of subsets of Ω. (...)*

1. *$P(A) \geq 0$ for all $A \in \mathbb{F}$.*
2. *$P(\Omega) = 1$.*
3. *If $A_1, A_2, \ldots$ is a collection of pairwise disjoint subsets of Ω that belong to F then are sets in $\mathbb{F}$ such that $A_i \subseteq A_{i+1}$ for all i, then $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$. (Pairwise disjoint $A_k \cap A_l = \phi$ if $k \neq l$)*

Properties:

1. *If $A \subseteq B$ then $P(A) \leq P(B)$. Proof: $B = A \cup (B \setminus A)$ and $P(B) = P(A) + P(B \setminus A)$ since A and $B \setminus A$ are disjoint. $P(B) = P(A) + P(B \setminus A) \geq P(A)$ since $P(B \setminus A) \geq 0$.*

Corollary 0.1. *Let $A \in \mathbb{F}$. (A is an event). Then $P(A) \leq 1$. Proof: $A \subseteq \Omega$, (...) by the previous proof, $P(A) \leq P(\Omega) = 1$.*

Proposition 4. *$P(A^c) = 1 - P(A)$. Proof: $\Omega = A \cup A^c$ and $A \cap A^c = \phi$. $P(\Omega) = P(A \cup A^c) = P(A) + P(A^c) = 1$, so $P(A^c) = 1 - P(A)$.*

Let A, B be events $(A, B \in \mathbb{F})$. Then $A \cap B \in \mathbb{F}$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. $A \cup B = A \cup (B \cap A^c)$, $A \cap (B \cap A^c) = \phi$, $P(A \cup B) = P(A) + P(B \cap A^c)$.

Lemma 1. *$P(B) = P(A^c \cap B) + P(A \cap B)$ and $P(A \cap B) = P(B) - P(A^c \cap B)$, $B = (A \cap B) \cup (A^c \cap B)$, $A \cap B$, $A^c \cap B$ are disjoint $P(B) = P(A \cap B) + P(A^c \cap B)$ and hence $P(A \cap B) = P(B) - P(A \cap B)$ and hence $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.*

The axioms define a probability, require THE countable additivity for piecewise disjoint sets IF $\{A\}_{i=1}^{\infty}$ are a collection of events such that $A_k \cap A_l = \phi$ if $k \neq l$ then $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$. If $\{A\}_{i=1}^{\infty}$ are not disjoint, how can we relate $P(\cup_{i=1}^{\infty} A_i)$ and $\sum_{i=1}^{\infty} P(A_i)$?

Proposition 5. *Let $\{A\}_{i=1}^{\infty}$ be a collection of events. Then $P(\cup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} P(A_i)$. Proof: define $A_i^{\star} = A_i$ and $A_2^{\star} = A_2 \setminus A_1$, $A_3^{\star} = A_3 \setminus (A_1 \cup A_2)$, $\ldots$, $A_n^{\star} = A_n \setminus (\cup_{i=1}^{n-1} A_i)$, $\ldots$, for $n \geq 2$.*

First, $\{A\}_{i=1}^{\infty}$ are pairwise disjoint. To show this, let $k \neq l$ be positive integers. We need to show that $A_k^\star \cap A_l^\star = \phi$. Assume that $l > k$ and suppose$A_k^\star \cap A_l^\star \neq \phi$. Let $x \in A_k^\star \cap A_l^\star$. Then $x \in A_l^\star$ and $x \in A_k^\star$. Since $x \in A_l^\star$ and $x \in A_k^\star$. Since $x \in A_l^\star$ then $x \in A_l$ but $x \notin \cup_{i=1}^{l-1} A_i$. Since $k < l$ then $x \notin A_k$. Futhermore, since $x \in A_k^\star$, then $x \in A_k$, we have here a contradiction. Hence $\{A_i^\star\}_{i=1}^{\infty}$ are piecewise disjoint. Then next step is to show that $\cup_{i=1}^{\infty} A_i = \cup_{i=1}^{\infty} A_i^\star$. First, note that since $A_i^\star \subset A_i$ for all $i \in \mathbb{N}$. Then $\cup_{i=1}^{\infty} A_i^\star \subset \cup_{i=1}^{\infty} A_i$. Now, we show that $\cup_{i=1}^{\infty} A_i \subset \cup_{i=1}^{\infty} A_i^\star$. Suppose $x \in \cup_{i=1}^{\infty} A_i$. Then, let i_0 be the smallest index integer such that $x \in A_{i_0}$. In particular, $x \notin A_1, x \notin A_2, \ldots, x \notin A_{i_0-1}$. Then $x \in A_{i_0}^\star \subset \cup_{i=1}^{\infty} A_i^\star$. Therefore, $\cup_{i=1}^{\infty} A_i^\star \subset \cup_{i=1}^{\infty} A_i^\star$. Then $\cup_{i=1}^{\infty} A_i^\star = \cup_{i=1}^{\infty} A_i$. $P(\cup_{i=1}^{\infty} A_i) = P(\cup_{i=1}^{\infty} A_i^\star) = \sum_{i=1}^{\infty} P(A_i^\star) \leq \sum_{i=1}^{\infty} P(A_i)$, since $A_i^\star \subset A_i$.

Corollary 1.1. *Let $A_1, A_2, \ldots, A_n$ be events. Then $P(\cap_{i=1}^{n} A_i) \geq \sum_{i=1}^{n} P(A_i) - (n-1)$. Proof: $P(\cap_{i=1}^{n} A_i) = 1 - P((\cap_{i=1}^{n} A_i)^c) = 1 - P(\cup_{i=1}^{n} A_i^c) \geq 1 - \sum_{i=1}^{n} P(A_i^c) = 1 - (n - \sum_{i=1}^{n} P(A_i)) = \sum_{i=1}^{n} P(A_i) - (n-1)$, since $P(\cup_{i=1}^{n} A_i^c) \leq \sum_{i=1}^{n} P(A_i^c)$.*

Problem 2. *Suppose $\{A_i\}_{i=1}^{\infty}$ are events such that $P(A_i) = 1$ for all $i \in \mathbb{N}$ such that $P(\cap_{i=1}^{\infty} A_i) = 1$.*[1]

Independent Events: Events A and B are said to be independent if $P(A \cap B) = P(A)P(B)$. (...) Finite sample spaces and equally likely outcomes

Example. *We roll a dce twice (tolls are assumed to be independent) and we write down the number obtained at each roll. The sample space $\Omega = \{(a,b), a, b \in \{1,2,3,4,5,6\}\}$. $|\Omega| = 36$. Since the two rolls are independent, the probability of $\{(a,b)\}$ is $1/6x1/6 = \frac{1}{36}$. The outcomes are equally likely. Let $\Omega = \{w_1, \ldots, w_n\}$ be a finite sample space with $|\Omega| = N$ we say that the outcomes $w_1, \ldots, w_N$ are equally likely if $P(\{w_1\}) = P(\{w_2\}) = \ldots = P(\{w_N\}) = \frac{1}{N}$. Let $A \subset \Omega$. Since Ω is finite, then A is finite and $A = \{w_1, w_2, \ldots, 2_d\}$ and $|A| = d$, $w_1, \ldots, w_n$ are the distinct d elements of A. $P(A) = \sum_{i=1}^{d} P(\{w_i\}) = \frac{1}{N} + \frac{1}{N} + \ldots + \frac{1}{N} = \frac{d}{N} = \frac{|A|}{|\Omega|}$.*

Example. *A fair coin is flipped twice. Suppose the flips are independent. The sample space $\Omega = \{HH, HT, TH, TT\}$ outcomes in Ω are equally likely with probability $1/4$.*

1. *Probability of getting at least one head, call this event A. Find P(A): $A = \{HH, HT, TH\}$, $|A| = 3$, $P(A) = \frac{3}{4}$.*

Counting

Theorem 2 (Fundamental Theorem of Counting)**.** *With m elements $a_1, \ldots, a_m$ and n elements $b_1, \ldots, b_n$, one can form $m \times n$ distinct pairs (a,b) where $a \in \{a_1, \ldots, a_m\}$ and $b \in \{b_1, \ldots, b_n\}$.*

[1] He'll prolly ask this on an exam.

Example. *how many 7 digit telephone numbers can one form? 10 chosen for first, 10 chosen for second, ..., 10 chosen for 7th How many 7-digit telephone numbers can one form if the third digit is even: 10 chosen for first, 10 chosen for second, 5 chosen for third, 10 chosen for fourth, ..., 10 chosen for seventh.*

Permutations

Example. *In a class of 100 students, 2 students are chosen. The first one wins 100$ and the second one wins 75$. How many choices are there if (1) a student can win twice (sampling with replacement) (2) a student can win at most once (sampling without replacement). (1) 100 choices for first, 100 choices for second, so* 100^2 *choices. (2) 100 choices for first, 99 choices for second, so* 100×99 *choices.*

Permutations: An ordered arrangement of r distinct objects is called a permutation. The number of ways of randomly choosing r ordered objects form n objects without replacement is denoted by P_r^n and is equal to $\frac{n!}{(n-r)!}$. In total, the number of ordering is $n \times (n-1) \times \ldots \times (n-r+1)$. $P_r^n = n \times (n-1) \times \ldots \times (n-r+1) = \frac{n!}{(n-r)!}$.

Example. *Two students are chosen to form a committee. In here many ways can the choice be done? 100 choices for first one, 99 choices for second one, 2! ways of ordering the two students, so* $\frac{100 \times 99}{2!}$ *ways.*

3 Appendix

4 Solutions

5 Useful Links